

Gold Future/ETF Analysis

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2025

New Data: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of 1, 2, 6 and 12-month futures on 1-day returns of spot gold.

Maturity	Slope	Intercept	R^2	RMSE
1-Month	0.9996	—	0.9494	—
2-Month	1.0024	—	0.9471	—
6-Month	1.0032	—	0.9407	—
12-Month	0.9478	—	0.8399	—

The slopes for the 1, 2, and 6-month contracts are all very close to 1, indicating a strong correlation with spot gold returns. The 12-month contract, however, has a noticeably lower slope of 0.9478 and a significantly lower R^2 of 0.8399. This may be a result of the lower liquidity and trading volume of the longer-dated contract, making it less sensitive to daily spot price movements.

Leung & Ward (2015) Data: Regression coefficients for one-day futures returns on one-day spot gold returns.

Maturity	Slope	Intercept	R^2	RMSE
1-Month	0.94314	2.4191×10^{-5}	0.80947	0.00530
2-Month	0.94301	1.0000×10^{-5}	0.80911	0.00530
6-Month	0.94348	3.1797×10^{-5}	0.80934	0.00530
12-Month	0.94358	-5.2333×10^{-5}	0.80984	0.00529

The slopes are also very close to 1, but consistently lower than the newer data. The R^2 values are also significantly lower than the newer data. This may be due to the smaller sample size of the older data and lower liquidity.

New Data: Slopes and R^2 from the regressions of futures returns versus gold returns over different holding periods.

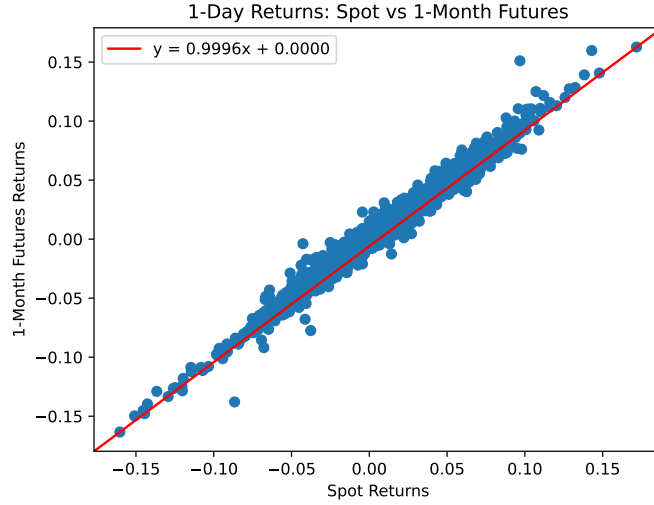


Figure 1: Scatter plot of one-day returns for spot gold versus 1-month gold futures. The regression slope of 0.9996 demonstrates a near one-to-one relationship.

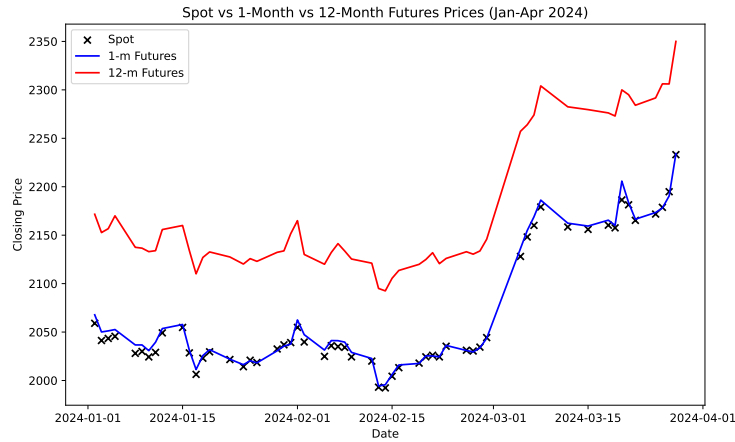


Figure 2: Comparison of closing prices for spot gold, 1-month futures, and 12-month futures in early 2024. The 1-month contract tracks the spot price closely, while the 12-month contract maintains a significant premium (contango).

	Days	1-Month	2-Month	6-Month	12-Month
Slope	1	0.9996	1.0024	1.0032	0.9478
	5	1.0074	1.0102	1.0106	0.9875
	10	1.0022	1.0040	1.0034	0.9897
	15	1.0031	1.0049	1.0033	0.9920
R^2	1	0.9494	0.9471	0.9407	0.8399
	5	0.9842	0.9840	0.9812	0.9567
	10	0.9915	0.9910	0.9878	0.9747
	15	0.9939	0.9933	0.9899	0.9807

The slopes and R^2 values generally increase with longer holding periods but decrease with longer-dated contracts.

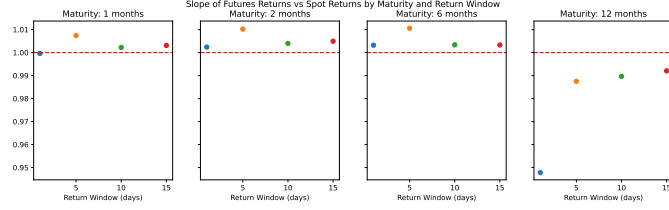


Figure 3: Regression slope of futures vs. spot returns, analyzed by contract maturity and the return window length in days. For most maturities, the slope approaches 1 as the return window lengthens.

Leung & Ward (2015) Data: Slopes and R^2 from regressions of futures returns vs. gold returns over different holding periods.

	Days	1-Month	2-Month	6-Month	12-Month
Slope	1	0.94314	0.94301	0.94348	0.94358
	5	0.99805	0.99752	0.99757	0.99715
	10	1.01075	1.01020	1.00970	1.00841
	15	0.99448	0.99461	0.99431	0.99345
R^2	1	0.80947	0.80911	0.80934	0.80984
	5	0.97154	0.97158	0.97194	0.97165
	10	0.98516	0.98530	0.98572	0.98550
	15	0.98704	0.98691	0.98725	0.98713

The out of sample period is defined as starting on January 1, 2024 and is approximately 10% of the data. The RMSEs are calculated in dollar terms assuming a notional investment of \$1000 in the spot gold price. The weights are calculated using a constrained least squares regression that forces the weights to sum to one. The GLD RMSE is \$6.00 for the out of sample period.

New Data: Weights and in/out of sample RMSEs for portfolios of 1, 2, 3, and 4 futures contracts.

Portfolio	w_0	w_1	w_2	w_6	w_{12}	RMSE _{in}	RMSE _{out}
1-m	0.00256	0.99743	—	—	—	\$2.65	\$5.15
2-m	0.00676	—	0.99324	—	—	\$3.67	\$5.75
12-m	0.04509	—	—	—	0.95491	\$18.84	\$17.95
1-m, 2-m	0.00168	1.21224	-0.21393	—	—	\$2.61	\$5.06
1-m, 6-m	0.00165	1.04753	—	-0.04918	—	\$2.61	\$4.88
1-m, 12-m	0.00189	1.01446	—	—	-0.01635	\$2.66	\$4.95
2-m, 6-m	0.00330	—	1.23075	-0.23405	—	\$3.18	\$4.63
2-m, 12-m	0.00310	—	1.09461	—	-0.09770	\$3.26	\$4.99
6-m, 12-m	0.00446	—	—	1.76090	-0.76537	\$6.07	\$12.15
1-m, 2-m, 6-m	0.00156	1.14088	-0.11333	-0.02911	—	\$2.60	\$4.94
1-m, 2-m, 12-m	0.00170	1.23840	-0.24373	—	0.00363	\$2.64	\$4.99
1-m, 6-m, 12-m	0.00197	1.10522	—	-0.17107	0.06388	\$2.62	\$4.54
2-m, 6-m, 12-m	0.00327	—	1.22084	-0.21809	-0.00602	\$3.21	\$4.46
1-m, 2-m, 6-m, 12-m	0.00186	1.21485	-0.13541	-0.14322	0.06191	\$2.62	\$4.62

The results show that using a single futures contract provides a reasonable replication, but performance diminishes as the maturity of the contract increases (out-of-sample RMSE rises from \$5.15 to \$17.95). Adding a second futures contract generally improves the out-of-sample RMSE, with the 2-month and 6-month combination performing best at \$4.63. The best-performing portfolio overall is the three-contract combination of 2-month, 6-month, and 12-month futures, which achieved an out-of-sample RMSE of \$4.46. Notably, several of these futures portfolios outperform the GLD ETF, which has an out-of-sample RMSE of \$6.00 over the same period.

Leung & Ward (2015) Data: Weights and in/out of sample RMSEs for portfolios of 1 and 2 futures contracts.

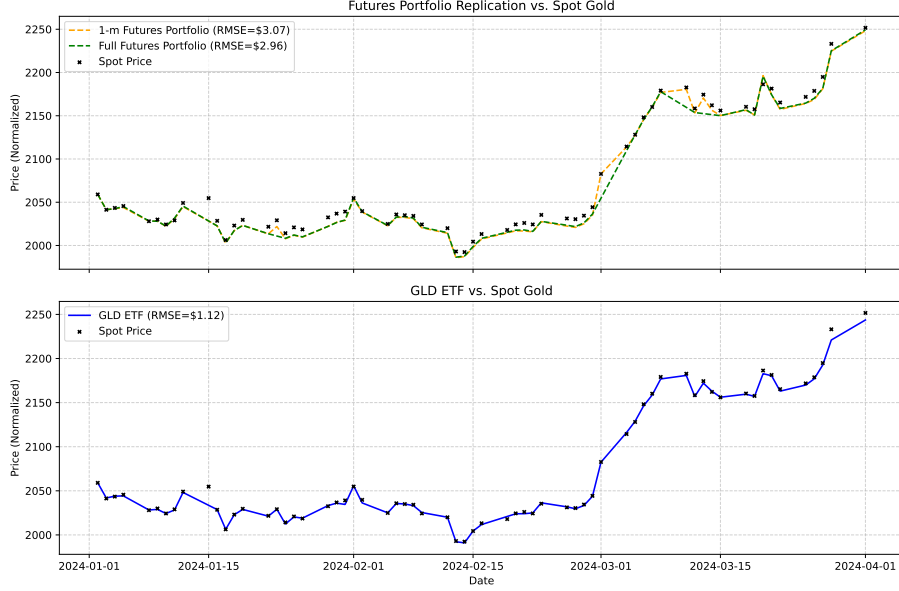


Figure 4: Visual comparison of tracking performance in early 2024. The top panel shows replication of spot gold using futures portfolios, while the bottom panel shows the GLD ETF’s tracking of spot gold. The futures portfolios achieve a lower RMSE in this period than GLD.

Portfolio	w_0	w_{short}	w_{long}	RMSE _{in}	RMSE _{out}
1-m	-0.01071	1.01071	—	\$6.63	\$3.34
2-m	-0.04835	1.04835	—	\$12.22	\$6.12
12-m	-0.06842	1.06842	—	\$15.11	\$5.71
1-m, 2-m	-0.00088	1.27315	-0.27227	\$6.27	\$2.79
1-m, 6-m	-0.00171	1.23899	-0.23729	\$6.27	\$2.98
1-m, 12-m	-0.00021	1.19336	-0.19315	\$6.29	\$3.00
2-m, 6-m	-0.04079	5.06602	-4.02523	\$10.27	\$9.37

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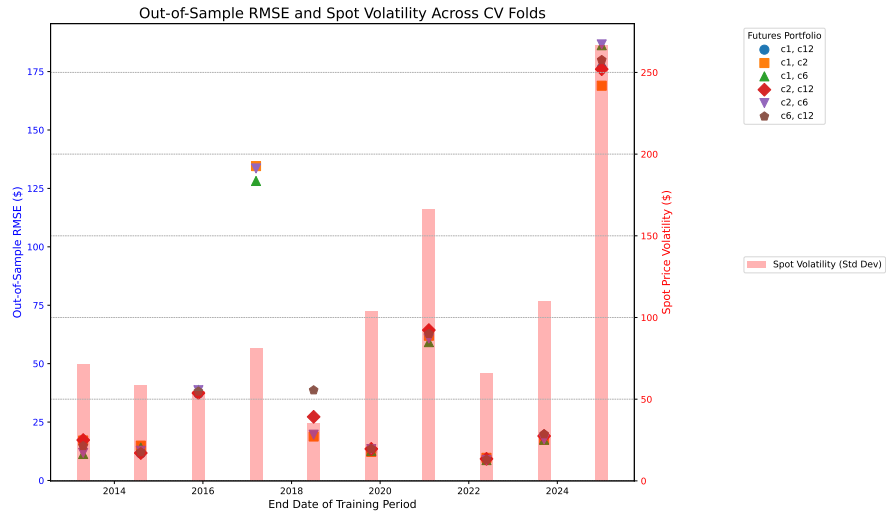


Figure 5: Out-of-sample RMSE for various futures portfolios across different cross-validation training periods. The pink bars represent the spot price volatility during each out-of-sample period. Tracking error tends to be higher in periods of higher market volatility, such as 2020 and post-2022.

New Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

(L)ETF	Leverage (β)	Slope	Intercept	t-stat	p-value	R^2	RMSE
GLD	1	1.0033	4.09×10^{-6}	1.440	0.150	0.9740	0.00178
UGL	2	2.0209	-1.95×10^{-4}	2.777	0.006	0.9644	0.00349
GLL	-2	-2.0143	1.89×10^{-4}	-1.923	0.055	0.9650	0.00345
DGLD	-2	-1.9756	-8.49×10^{-5}	0.468	0.640	0.4066	0.02098
UGLDF	3	3.0063	4.59×10^{-4}	0.058	0.954	0.2552	0.04463

The regression results indicate that GLD, UGL, and GLL are effective at tracking their underlying benchmarks, with high R^2 values and slopes very close to their target betas. The p-value for GLD (0.150) shows its slope is not statistically different from 1. However, the p-values for UGL (0.006) and GLL (0.055) suggest their slopes are statistically different from their target betas of 2 and -2, respectively. DGLD and UGLDF perform poorly, with very low R^2 values (0.4066 and 0.2552), indicating they do not reliably track their leveraged benchmarks.

Leung & Ward (2015) Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

(L)ETF	Leverage (β)	Slope	Intercept	t-stat	p-value	R^2	RMSE
GLD	1	1.00540	-1.6427×10^{-6}	1.42692	0.15382	0.98060	0.00163
UGL	2	2.00572	-1.3107×10^{-4}	0.73319	0.46357	0.97930	0.00337
GLL	-2	-2.00556	-1.0550×10^{-4}	0.67089	0.50240	0.97673	0.00358
UGLD	3	2.99358	-1.9860×10^{-4}	0.45211	0.65133	0.98484	0.00421
DGLD	-3	-2.97528	-1.6912×10^{-5}	0.97442	0.33019	0.95256	0.00752

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New Data: Weights and in/out of sample RMSEs for portfolios replicating leveraged benchmarks.

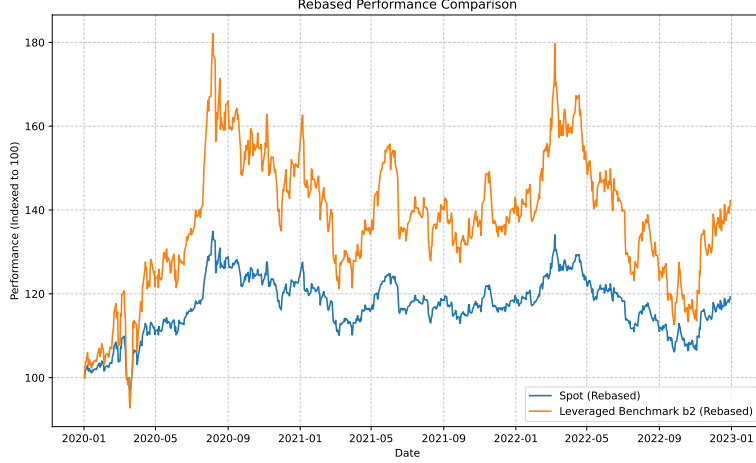


Figure 6: Performance of spot gold versus a 2x leveraged benchmark from 2020 to 2023, with both series indexed to 100. This illustrates the compounding effect (volatility drag) over long periods, where the leveraged product’s return is not simply double the spot return.

Benchmark	Portfolio	w_0	w_1	w_2	w_6	w_{12}	RMSE_{in}	RMSE_{out}
2x Leveraged	1-m	-1.59664	2.59797	—	—	—	\$53.36	\$322.86
	2-m	-1.58673	—	2.58801	—	—	\$50.91	\$325.15
	6-m	-1.54780	—	—	2.54977	—	\$42.93	\$342.68
	12-m	-1.47123	—	—	—	2.47233	\$44.81	\$345.98
	1-m, 2-m	-1.54673	-9.50324	12.05135	—	—	\$46.67	\$333.81
	1-m, 12-m	-1.51211	0.92136	—	—	1.59137	\$41.50	\$335.50
3x Leveraged	1-m	-3.96685	4.96535	—	—	—	\$175.90	\$1011.78
	2-m	-3.94914	—	4.94744	—	—	\$171.36	\$1015.26
	6-m	-3.88232	—	—	4.88107	—	\$153.51	\$1039.95
	12-m	-3.72756	—	—	—	4.72473	\$143.57	\$1033.75
	1-m, 2-m	-3.81349	-32.24312	37.05523	—	—	\$156.89	\$1038.14
	1-m, 12-m	-3.71268	-0.33417	—	—	5.04421	\$143.45	\$1036.52

Static replication performs poorly when attempting to track leveraged benchmarks. While in-sample RMSEs are already significantly higher than for the spot benchmark, the out-of-sample RMSEs are extremely large (e.g., \$322.86 for a single 1-month future replicating a 2x benchmark, and \$1011.78 for a 3x benchmark). This demonstrates that a static portfolio of futures cannot capture the path-dependent nature and daily compounding effects inherent in leveraged ETFs, leading to large tracking errors over time.

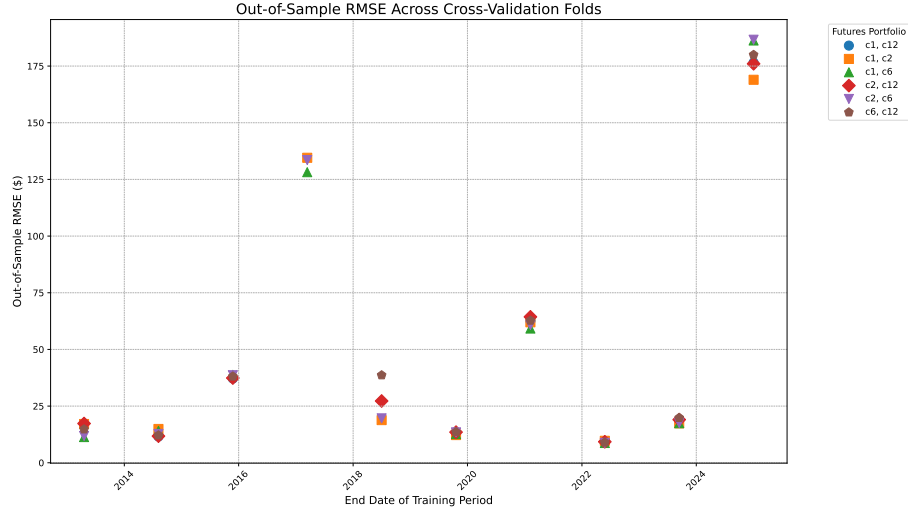


Figure 7: Out-of-sample RMSE for various futures portfolios attempting to replicate a 2x leveraged benchmark across different cross-validation folds. The tracking error is consistently high across all time periods, though it varies with market conditions.

Leung & Ward (2015) Data: RMSEs for portfolios replicating leveraged benchmarks.

Benchmark	Portfolio	RMSE _{in}	RMSE _{out}
+2x (UGL)	1-m	\$153.20	\$41.49
	2-m	\$140.41	\$49.57
	6-m	\$138.63	\$47.46
	12-m	\$134.51	\$48.30
+3x (UGLD)	1-m	\$555.49	\$111.51
	2-m	\$529.91	\$125.99
	6-m	\$526.58	\$122.33
	12-m	\$518.97	\$123.83

1 Dynamic Replication

Out-of-sample RMSE for dynamically replicated portfolios versus the actual (L)ETF's tracking error.

(L)ETF	Replicated RMSE by Futures Maturity				Actual ETF RMSE
	1-Month	2-Month	6-Month	12-Month	
GLD	\$113.94	\$117.11	\$120.92	\$288.55	\$55.79
UGL	\$816.20	\$825.18	\$861.66	\$907.40	\$176.72
GLL	\$1120.26	\$1120.72	\$1124.66	\$1050.30	\$1094.66
DGLD	\$805.47	\$805.11	\$809.21	\$773.20	\$896.73
UGLDF	\$669.66	\$644.20	\$669.53	\$1196.78	\$198.11

The performance of dynamic replication, which involves daily rebalancing of a single futures contract, varies significantly across different (L)ETFs. For positively leveraged ETFs like UGL (2x) and UGLDF (3x), as well as the 1x GLD, the dynamic replication strategy results in a substantially higher tracking error (RMSE) than the actual ETFs. In contrast, for the negatively leveraged GLL (-2x), the replicated portfolio's RMSE is very close to that of the ETF itself. Notably, for DGLD (-2x), the dynamically replicated portfolios, particularly the one using the 12-month future, achieve a lower RMSE (\$773.20) than the actual ETF (\$896.73), indicating a superior tracking performance by the model in this specific case. This suggests that while the dynamic replication model is a useful theoretical framework, its practical effectiveness depends heavily on the specific characteristics and tracking efficiency of the target (L)ETF.